Artificiall Intelligence

2019 - Midterm 1 - Spring

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BDS - 6A

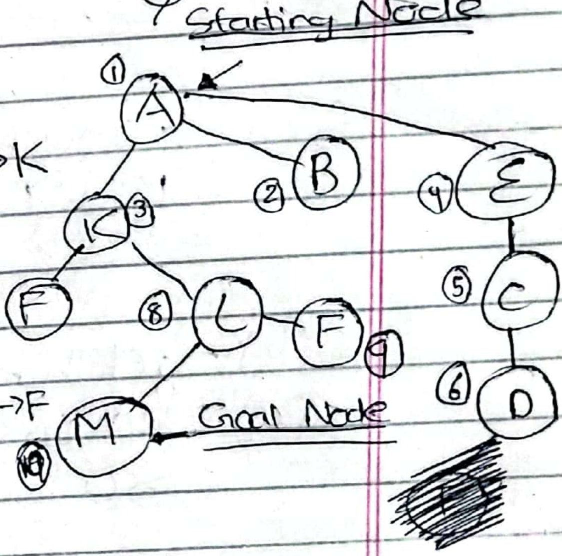
Question 1

(a)

(i)

① $f(A) = g(A) + h(A) = 0 + 56 = 56$ } A

② $f(B) = g(B) + h(B) = 30 + 22 = 52$ } A->B



③ $f(K) = g(K) + h(K) = 20 + 30 = 50$ } A->K

⑧ $f(F) = g(F) + h(F) = 20+17+30 = 67$ } A->K->F

⑦ $f(L) = g(L) + h(L) = 20+15+15 = 50$ } A->K->L

⑨ $f(M) = g(M) + h(M) = 20+15+15+0 = 50$ } A->K->L->M

④ $f(E) = g(E) + h(E) = 12 + 29 = 41$ } A->E

$f(F) = g(F) + h(F) = 20+15+10+30 = 75$

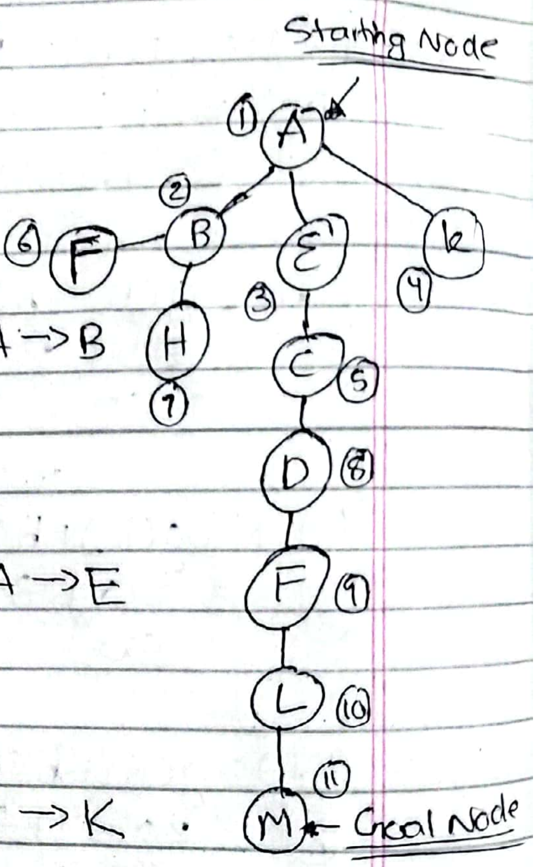
⑤ $f(C) = g(C) + h(C) = 12+5+30 = 47$ } A->E->C

(6) $f(D) = g(D) + h(D)$
$= 12 + 5 + 5 + 29$
$= 51$
} A -> E -> C -> D

~~f(F) = g(F) + h(F) = 12 + 5 + 5 + 5 + 30 = 57 } A -> E -> C -> D -> F~~

iii) Path = A -> K -> L -> M

(b) (i)



(1) $f(A) = g(A) + h(A)$
$= 0 + 80$
$= 80$
} A

(2) $f(B) = g(B) + h(B)$
$= 30 + 10$
$= 40$
} A -> B

(3) $f(E) = g(E) + h(E)$
$= 12 + 10$
$= 22$
} A -> E

(4) $f(k) = g(k) + h(k)$
$= 20 + 60$
$= 80$
} A -> K

(5) $f(C) = g(C) + h(C)$
$= 12 + 5 + 50$
$= 67$
} A -> E -> C

(6) $f(H) = g(H) + h(H)$
$= 30 + 11 + 30$
$= 71$
} A -> B -> H

$f(F) = g(F) + h(F) = 30 + 19 + 30 = 79$ } A -> B -> F

$f(D) = g(D) + h(D)$
⑦ $= 12+5+5+20$ } $A \to E \to C \to D$
$= 42$

$f(F) = g(F) + h(F)$
⑧ $= 12+5+5+5+30$ } $A \to E \to C \to D \to F$
$= 57$

~~f(B) = g(B) + h(B) = 12+5+5+5+19 = 46~~

$f(L) = g(L) + h(L)$
⑨ $= 12+5+5+5+10+20$ } $A \to E \to C \to D \to F \to L$
$= 57$

$f(M) = g(M) + h(M)$
⑩ $= 12+5+5+5+10+15+0$ } $A \to E \to C \to D \to F \to L \to M$
$= 52$

iii) Path $\Rightarrow A \to E \to C \to D \to F \to L \to M$

(c) The optimality of the runs depends on wether the heuristic function is admissible or not. An admissible heuristic never overestimates the cost to reach the goal. In both cases, the heuristic functions provided are admissible as they are straight-line distances, which are always less than or equal to the actual distances.

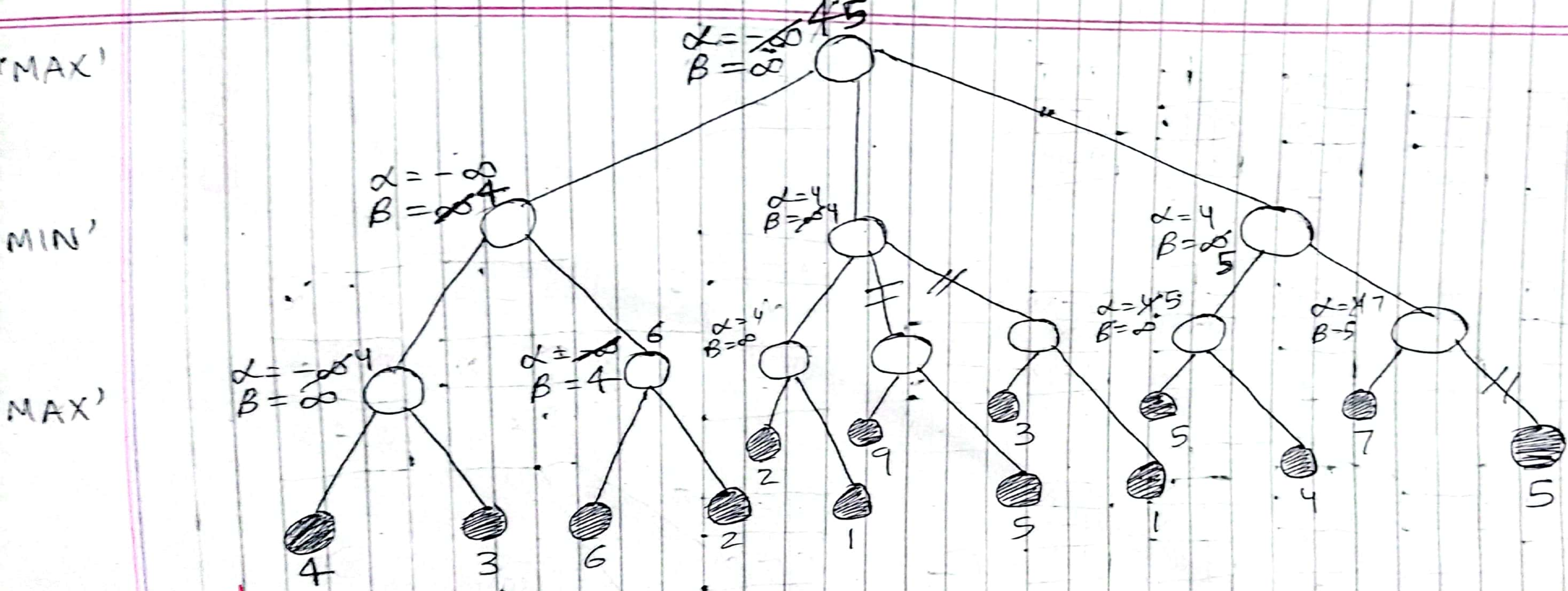
The difference in routes returned by the two runs is due to the difference in the heuristic functions used. In the first run, the heuristic function provides more accurate estimates of the distances

MAX) MIN) (MAX)

...tion 2

to the goal, leading to a more direct path. In the second run, the heuristic function is less accurate, resulting in a longer path to the goal.

~~Question 2~~ Time Taken 90 minutes



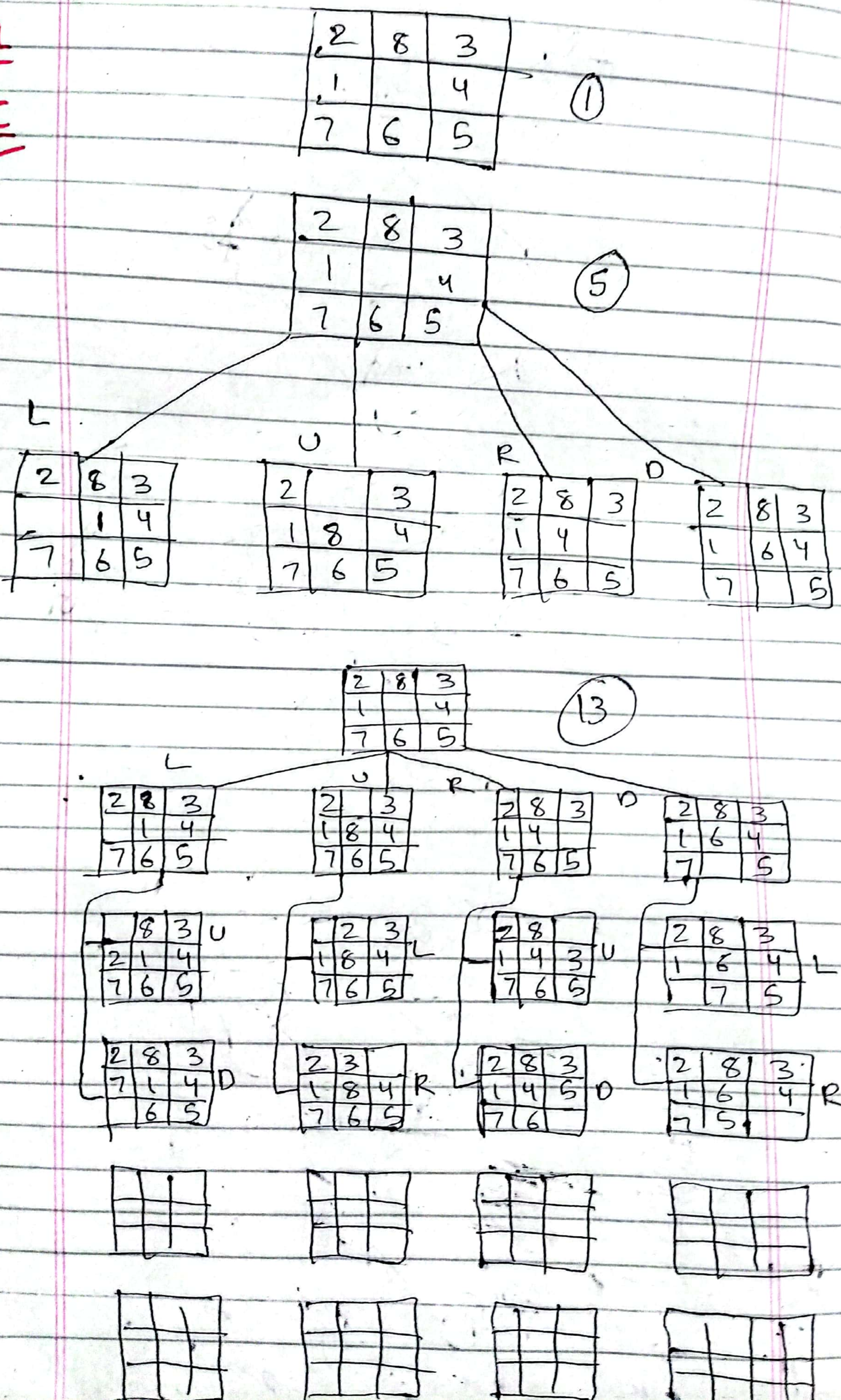
Question 2

Time taken = 45 mins

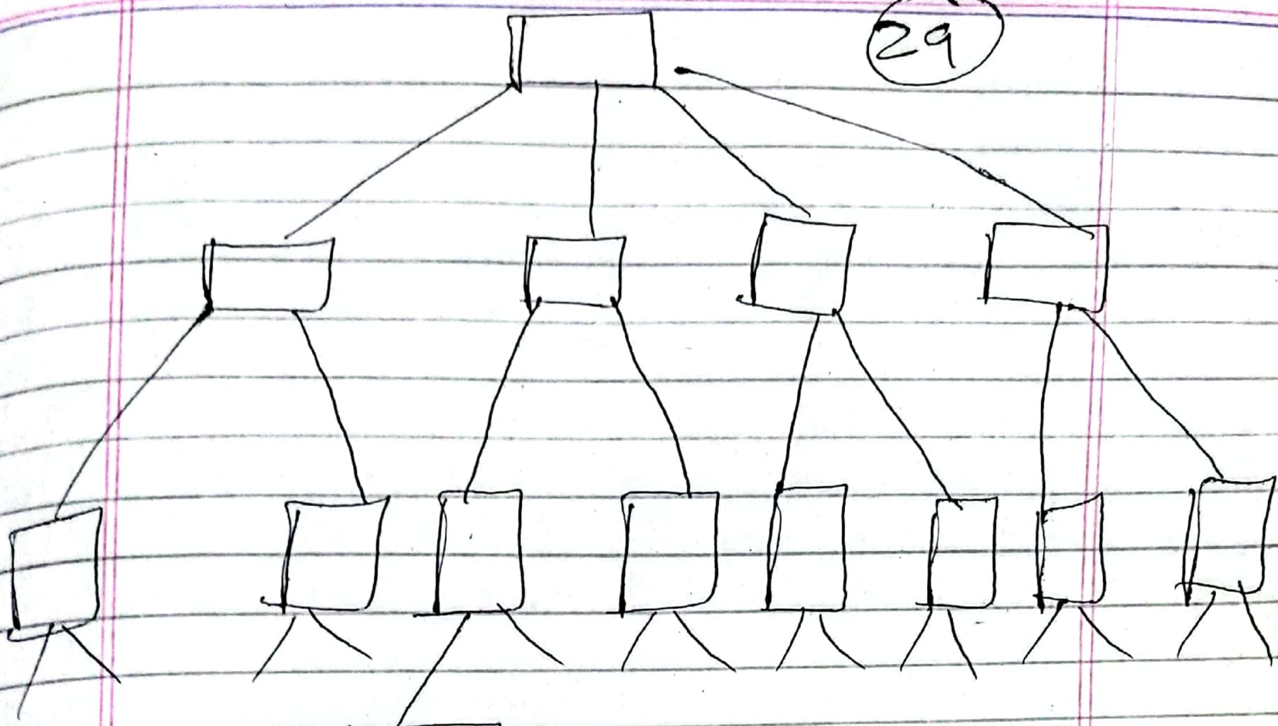
Question 3

a

c



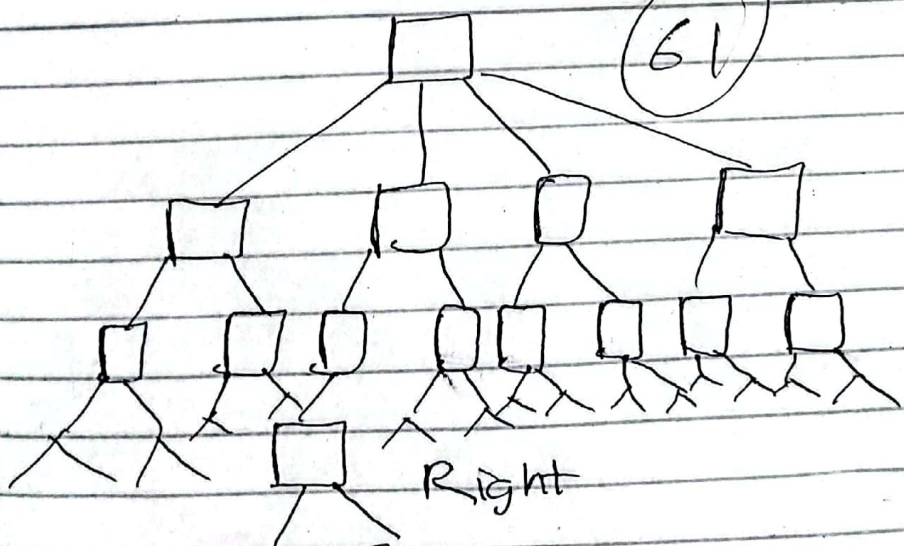
29



1	2	3
	8	4
7	6	5

Down

61



Right

1	2	3
8		4
7	6	5

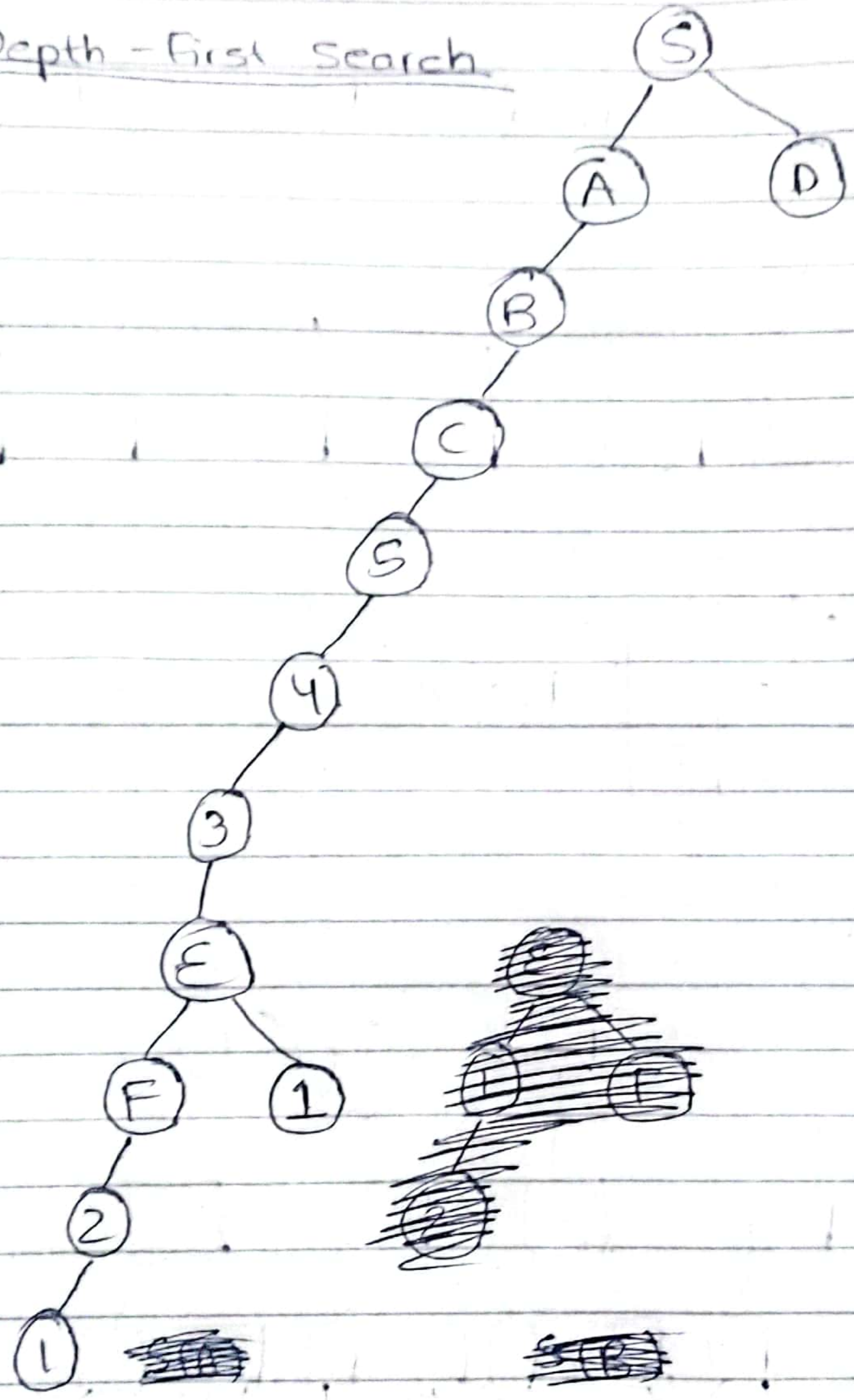
b Path = U, L, D, R

Time Taken = 35 min

Spring 2023

Question 1

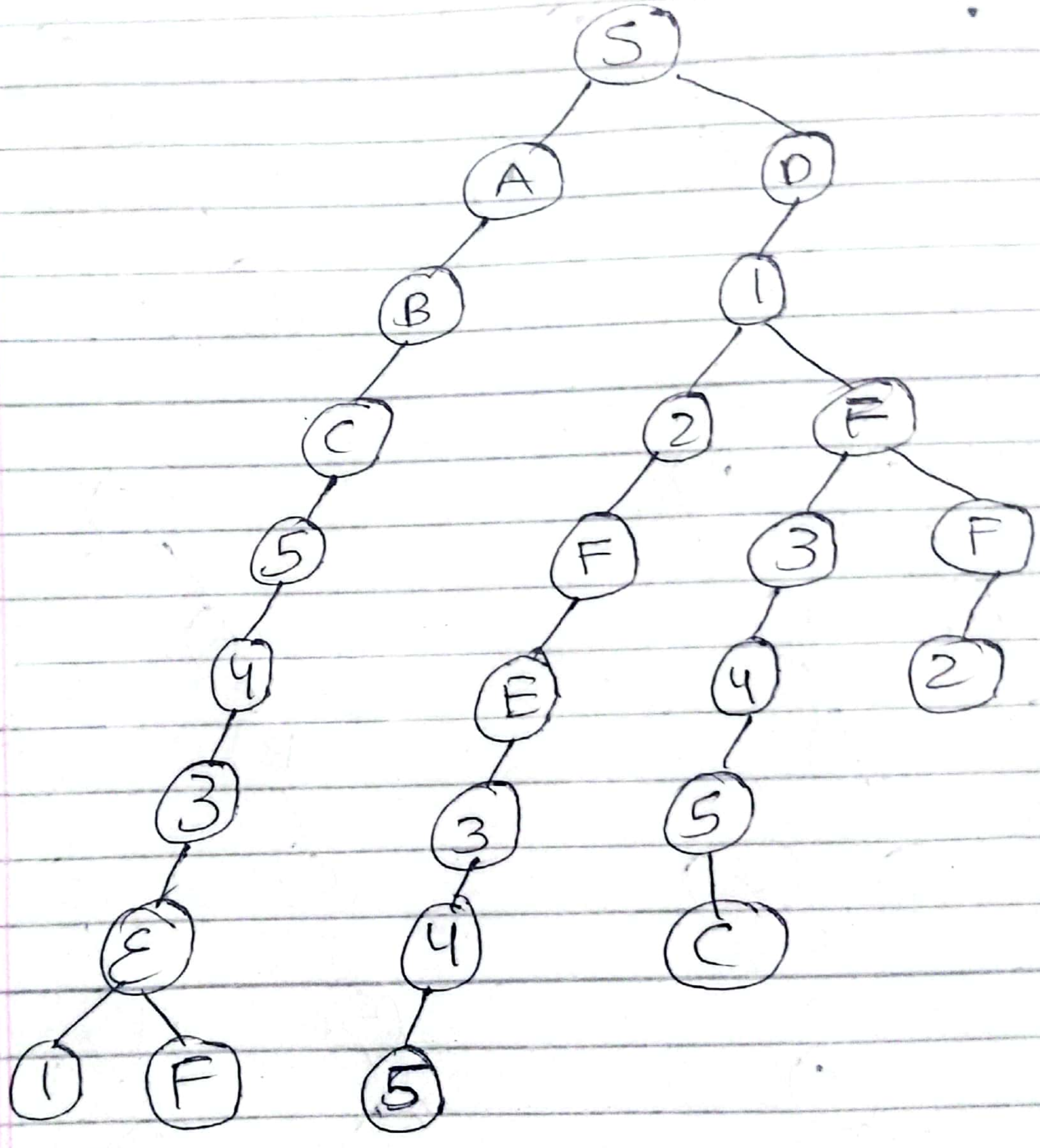
a. Depth - First Search



Path length = 10 Nodes = 13

Path = $S \rightarrow A \rightarrow B \rightarrow C \rightarrow 5 \rightarrow 4 \rightarrow 3$
$\rightarrow E \rightarrow F \rightarrow 2 \rightarrow 1$

Breadth - First Search

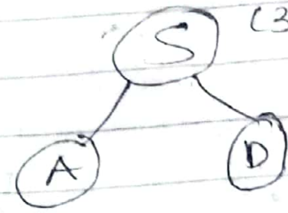


Path = S → D → 1 → 2 → F → E → 3 → 4 → 5

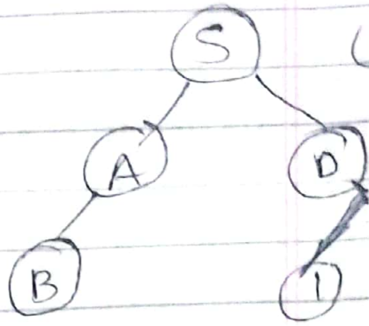
Length = 8
Nodes = 26

Iterative Deepening Search

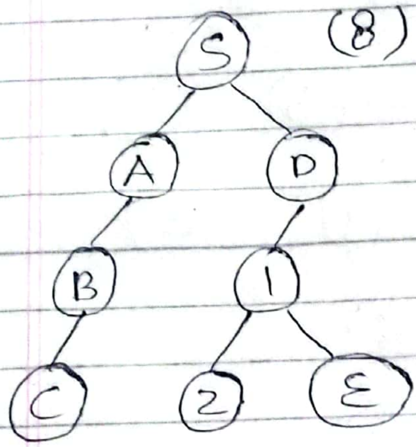
(1)



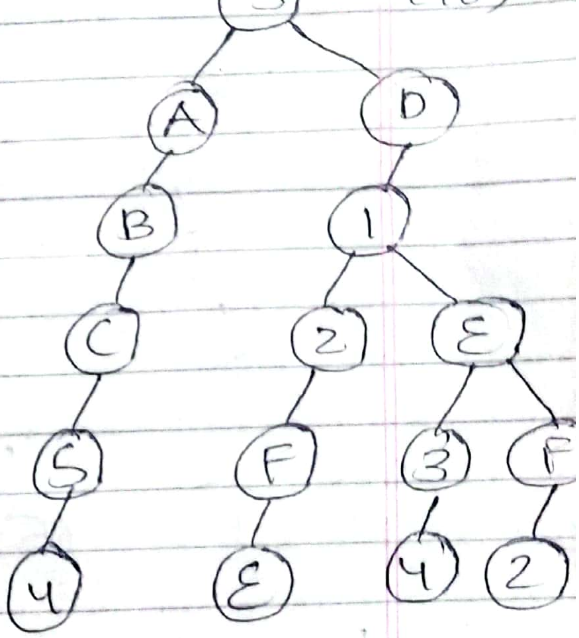
(3)



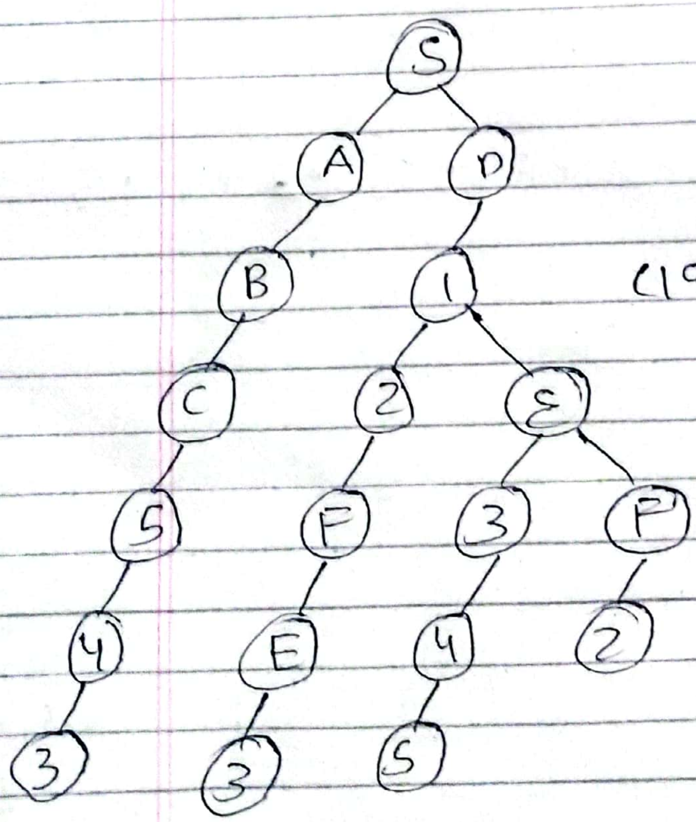
(5)



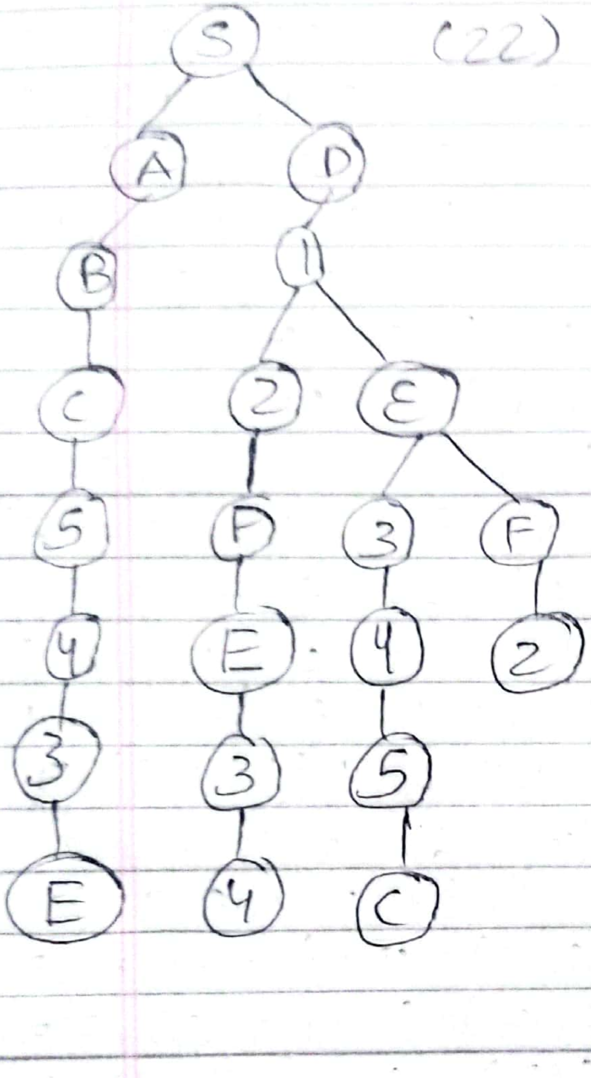
(8)

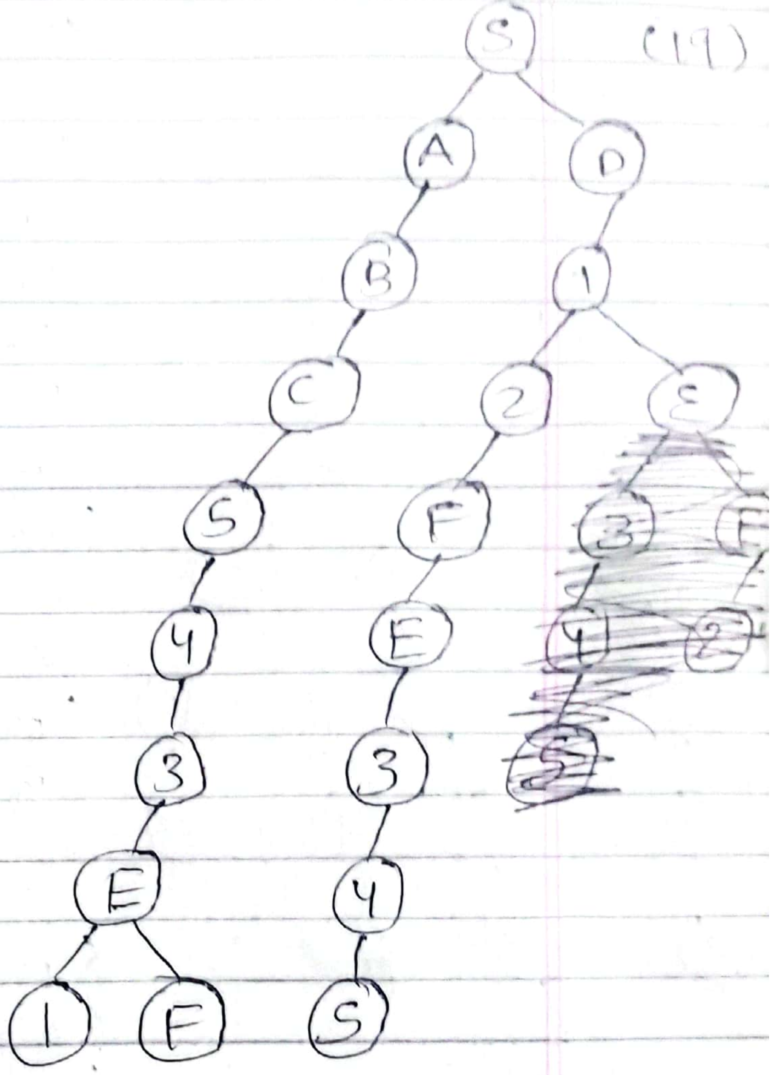


(16)



(19)



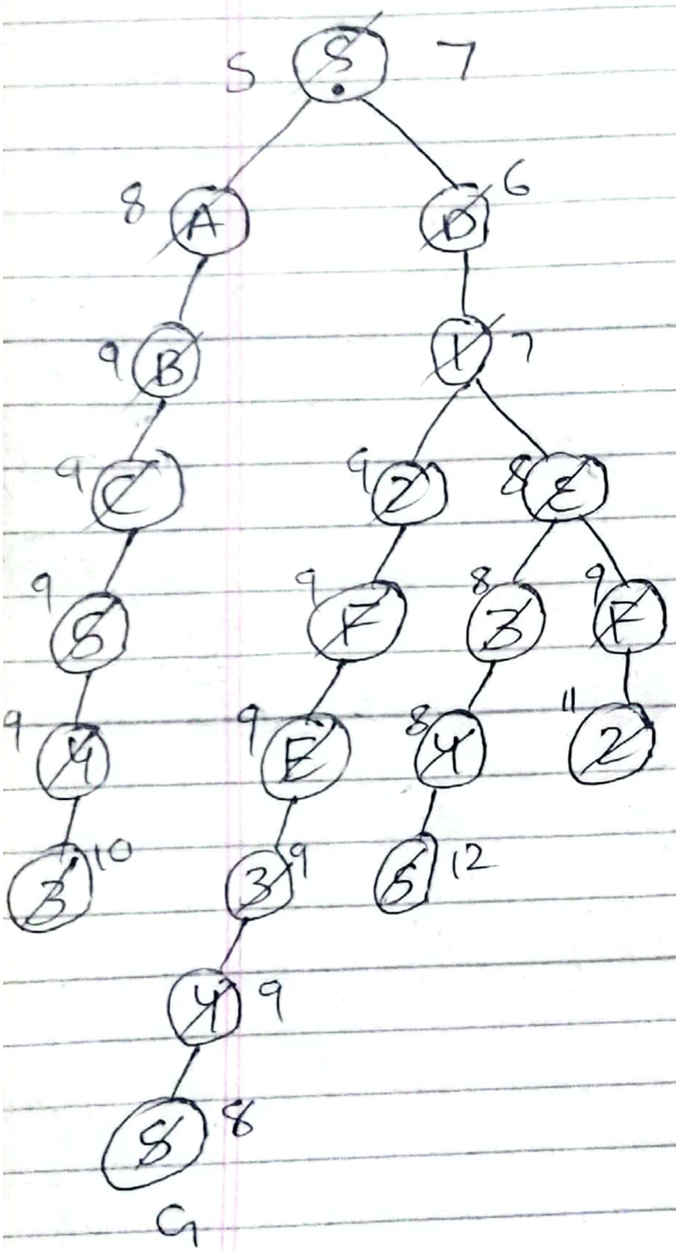


Nodes = 105

Path = S → D → 1 → 2 → F → E → 3 → 4 → 5

length = 8

(b) A* Search



$X v_1 \; f(S) = 0+(2,3,4,5,4)+5 = \cancel{7}$

$X v_4 \; f(A) = 1+(3,4,3,4,3)+5 = \cancel{8}$

$X v_2 \; f(D) = 1+(1,2,3,4,5)+5 = \cancel{6}$

$X v_3 \; f(1) = 2+(1,3,4,5)+4 = \cancel{7}$

$X v_{11} \; f(2) = 3+(3,4,5)+3 = \cancel{9}$

$X v_4 \; f(E) = 3+(1,2,3)+4 = \cancel{8}$

$X v_7 \; f(B) = 2+(4,5,2,3,2)+5 = \cancel{9}$

$X v_5 \; f(3) = 4+(3,1,2)+3 = \cancel{8}$

$X v_{10} \; f(4) = 5+(1,4)+2 = \cancel{8}$

$f(5) = 6+(5)+1 = 12$

$X v_7 \; f(C) = 3+(6,5,3,2,1)+5 = \cancel{9}$

$X v_8 \; f(5) = 4+(5,4,2,1)+4 = \cancel{9}$

$X v_9 \; f(4) = 5+(3,4,1)+3 = \cancel{9}$

$f(3) = 6+(2,3)+2 = 10$

$f(2) = 5+(3,4,5)+3 = 11$

$X v_{12} \; f(F) = 4+(2,3,4)+3 = \cancel{9}$

$X v_{13} \; f(E) = 5+(1,2,3)+3 = \cancel{9}$

$X v_{14} \; f(3) = 6+(1,2)+2 = \cancel{9}$

$X v_{15} \; f(4) = 7+(1)+1 = \cancel{9}$

$X v_{16} \; f(5) = 8+(0)+0 = \cancel{8}$

Path = S → D → 1 → 2 → F → E → 3 → 4 → 5

Path Length = 8

Optimal = Yes

Nodes = 21

C ① c) Greedy best first search with admissible heuristics

② d) All of the above

③ Completeness is that the search always find a solution if one exists while optimality is that search always finds a least-cost solution.

④ Yes, the heuristic function is admissible because the actual cost to reach a dot is greater than or equal to the estimated heuristic value of H(n). Actual cost is greater because it incorporates the cost of hurdles placed in the path but H(n) is always underestimating this value.

⑤

Algorithms	DFS	BFS	IDS	A*
Total No. of nodes	13	25	105	21
Solution path found	SABC5943E F21	SD12FE 345	SD12FE 345	SD12FE 345
Solution path length	10	8	8	8
Optimal	No	Yes	Yes	Yes